

INTRODUCTION

The problem of finding the square root of a graph is a well-studied field in both mathematics and computer-science [MS94] and [FK12], with specific relation to distributed computing, network design and computational biology. Research in this area has primarily focused around the undirected graph H and its square $G = H^2$ which is obtained by adding an edge between every pair of vertices whose distance in H is at most two.

BACKGROUND

We examine the relaxed linear program which can be evaluated in polynomial time using interior point method, and integrate existing literature on special root structures as a stepping-stone towards generating a true polynomial time algorithm for subclasses of graphs.

Let the *graph root* $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. Denote by $d_H(u, v)$ the *graph distance* defined as the number of edges of the shortest path between vertices u and v in H . If no such path exists, $d_H(u, v) = \infty$.

The *square* of the graph H , denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \quad (1)$$

That is, two distinct vertices are adjacent in H^2 if and only if they are connected by a path of length one or two in H .

Using an alternative construction, the same vertex set V . Let $A_H \in \{0, 1\}^{n \times n}$ denote the adjacency matrix of H , and let $D_H \in \mathbb{R}^{n \times n}$ denote the degree matrix of H . Let $\mathbb{1}_{\geq 1} : \mathbb{R}^{n \times n} \rightarrow \{0, 1\}^{n \times n}$ denote the entrywise *indicator function*, defined by

$$\mathbb{1}_{\geq 1}(M)_{ij} = \begin{cases} 1, & \text{if } M_{ij} \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

Define the adjacency matrix of G using the adjacency and degree matrices of H

$$A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \quad (2)$$

The graph G defined by (2) is equivalent to the graph square H^2 defined in (1)

$$ij \in E_{H^2} \iff 1 \leq d_H(i, j) \leq 2. \quad (3)$$

AIM

Much of the existing research on graph square roots has centered on characterizing when a graph admits such a root and on the complexity of reconstructing H from G . In contrast, rather than studying the construction problem directly, we analyze the relationship between the adjacency matrix A_H of the root and the adjacency matrix A_G of the square, and show that the problem can be completely categorized as an integer program.

LINEAR PROGRAMMING

The formulation of the adjacency matrix of the graph square in Equation (2) is very powerful and gives the integer program

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (4)$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (5)$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (6)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (7)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (8)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (9)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (10)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (11)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (12)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (13)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (14)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (15)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (16)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (17)$$

The linear relaxation of the integer program defined by Equations (4)–(17) is constructed in order to generate a feasible weighted solution in polynomial time algorithm. It is done with identical constraints, however the decision variables (4), (5) and (6) are replaced with

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (18)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (19)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (20)$$

SINGLE TRIANGULAR DECOMPOSITION

Using the local triangular structure of the graph, certain edge assignments in a square root can be fixed. For a vertex $s \in V$, define

$$T(s) = \{\{u, v\} \subseteq V \setminus \{s\} \mid (s, u) \in E, (s, v) \in E, (u, v) \in E\}$$

to be the set of 3-cliques or triangles in G that contain s . If $|T(s)| = 1$ where $T(s) = \{\{u, v\}\}$ then the vertex s exists in a unique triangle of G . When the neighbours are structurally distinct with different degree $\deg_G(u) \neq \deg_G(v)$, then the vertex with smaller degree must be the stem vertex in the square root. This is assigned deterministically using

$$x_{sw} = 1, \quad x_{ws} = 1 \text{ where } w = \arg \min\{\deg_G(u), \deg_G(v)\}. \quad (21)$$

Ross and Harary in 1960 gave a polynomial time algorithm for constructing a tree root by using characterizations of the square graph to conclude that existing tree square roots are unique up to isomorphism [RH60].

Proposition 1. *If a square graph G has a tree root H , then the single triangular decomposition linear program returns an integral solution.*

Proof. Assume for contradiction that there exists a square graph G with a triangle root H such that the linear program has an optimal solution x^* that is fractional. Then there exists at least one edge $ij \in E_G$ such that $0 < x_{ij}^* < 1$.

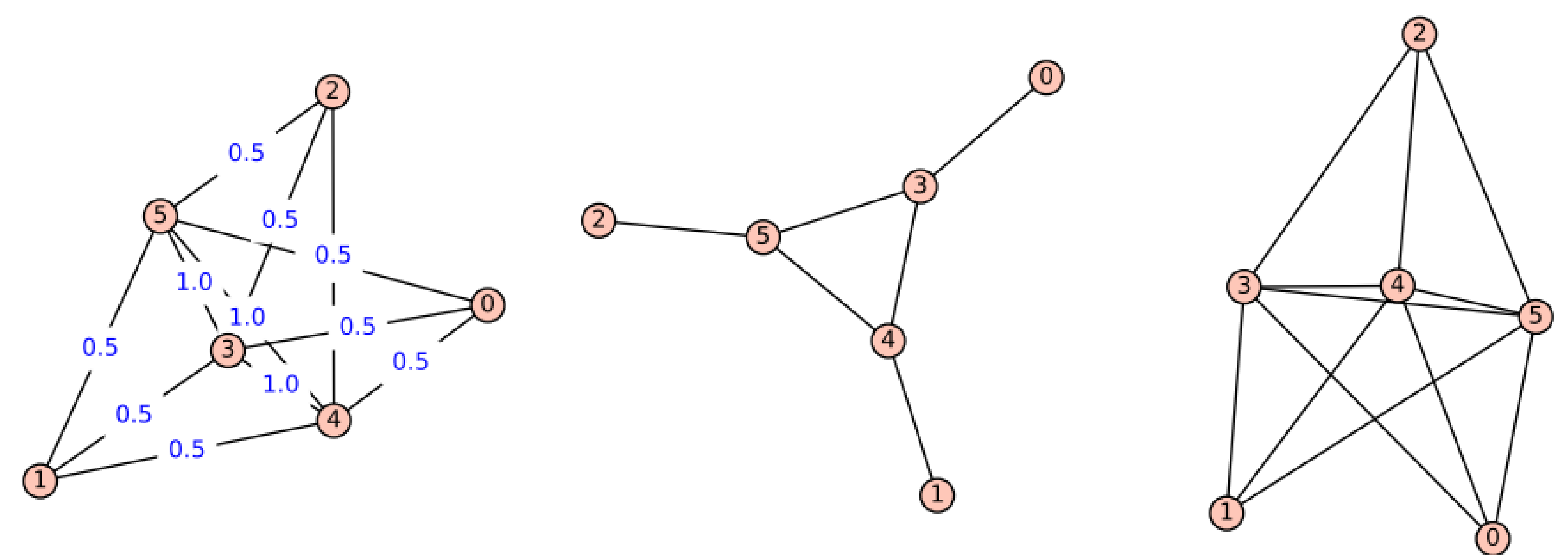
If G has a vertex v that is only in one triangle, then v must be a pendant in H . As the linear program applies (21), any pendant edge will be integral. Hence, any fractional edge must be incident only to vertices of degree at least two in the support of x^* .

Consider such a fractional edge ij with endpoints i, j . Since both endpoints are not pendants, each is incident to at least one other edge with positive weight. Iterating this argument yields a maximal connected fractional root consisting entirely of edges with endpoints that are not pendants.

If all vertices G are in more than one triangle, then there are no pendant vertices in H and hence all vertices in the root are either leaves or internal vertex. However, if the root has two leaves coming off of the same internal vertex, then an additional edge between the two leaves can be added such that the square is the same. Indeed, this new root is not only feasible but also has greater weight and hence would have been chosen by the linear program to be the optimal solution.

This implies that no internal vertex may support two leaves and hence that the root cannot branch. Therefore, the fractional root must be a simple path. $\square$

EXAMPLE



Example of a fractional linear program solution, its root and the original square.

REFERENCES

- [FK12] Babak Farzad and Majid Karimi. “Square-Root Finding Problem In Graphs, A Complete Dichotomy Theorem”. In: (2012).
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- [RH60] Ian C. Ross and Frank Harary. “The square of a tree”. In: *The Bell System Technical Journal* 39.3 (1960), pp. 641–647.